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DATE: 09/03/2026

STATUS: APPROVED

NLX2500 SS Tool Post Calibration

Equipment / System No _____ Date _____ Performed by _____

SAFETY / ENVIRONMENTAL

Verify that all personnel who will be involved with this activity are familiar with this procedure, any risks or hazards that may be encountered, possess the required safety equipment, and are familiar with the site safety and environmental policies.

OBJECTIVE

The purpose of this procedure is to correctly calibrate the sub-spindle tool setter (removable tool-eye arm) on the Mori Seiki NLX2500SY/700 after the tool setter disc/contact post has been replaced, by re-teaching the sensor-position parameters (2097X / 2098X / 2097Z / 2098Z) so that tools measured on the sub-spindle read true.

OVERVIEW

Replacing the tool setter disc/contact post physically relocates the four measuring surfaces of the sub-spindle tool-eye. Until the stored sensor-position parameters are re-taught, every tool measured on the sub-spindle side will be off by the amount the post moved. This procedure re-establishes those parameters using a mic-verified reference tool and a measure-error to adjust-parameter loop, then verifies to tolerance. Perform with SINGLE BLOCK enabled and rapid/feed override turned down.

PROCEDURE

Task 1 - Preparation and safety

□ Confirm the new tool setter disc/contact post is fully seated and torqued on the removable sub-spindle arm.

□ Enable SINGLE BLOCK; turn rapid override and feed override down. Keep one hand near FEED HOLD throughout.

□ On the sensor-position parameter screen, RECORD the current values of 2097X, 2098X, 2097Z, and 2098Z on paper so they can be reverted if needed.

□ Deploy the removable sub-spindle tool-eye arm and confirm it is seated and clear of the turret path.

Task 2 - Establish a reference (truth) tool

□ Chuck a test bar in the SUB-SPINDLE.

□ Face the bar; set the reference tool's Z geometry offset to the faced surface.

□ Turn the OD a short distance, retract without moving X, and measure the diameter with a micrometer.

□ Set the reference tool's X geometry offset from the measured diameter. Mic-verify. This tool is now the calibration reference.

Task 3 - Measure the reference tool on the setter

□ With the reference tool selected, touch it off on the sub-spindle setter using the normal tool-measurement function.

□ Write down the X (and Z) geometry value the setter reports for the reference tool.

Task 4 - Cut a test part and calculate the error

□ With the reference tool, take a skim finish cut on the sub-spindle part.

□ Micrometer the actual cut DIAMETER (X) and the actual FACE/length (Z).

□ X error = setter-reported diameter minus mic'd diameter. Convert to millimeters if needed, then DIVIDE BY 2 to get the radial correction.

□ Z error = setter-reported face minus mic'd face (direct value, no halving).

Task 5 - Correct the sensor-position parameters

□ Navigate to the sub-spindle tool-eye sensor-position parameters per the machine book (SYSTEM to SETUP to AXIS, page to the 209x block).

□ X axis: shift BOTH 2097X and 2098X by the Task 4 radial value. If the test part measured OVERSIZE add the value; if UNDERSIZE subtract it (verify this sign once on this machine, then record it on this sheet).

□ Z axis: shift BOTH 2097Z and 2098Z by the Task 4 Z error, using the same oversize/undersize sign logic.

□ Move both members of each axis pair by the SAME amount - you are translating the re-installed post, not changing its geometry.

Task 6 - Verify and iterate

□ Repeat Task 3 and Task 4. Because the entire post was replaced, expect 2 to 3 iterations to converge.

□ Continue until the setter reading agrees with the mic within plus/minus 0.0002 in (plus/minus 0.005 mm) on both axes.

□ Confirm on a second tool plus a test cut before trusting production offsets.

□ If X and Z are both far off after one pass, a sign is likely flipped - revert to the recorded values and redo that axis. If it will not converge, the arm is not repeatably seating (mechanical issue), not a parameter issue.

□ Record the final 2097X / 2098X / 2097Z / 2098Z values and the verified X sign direction on this sheet.

Completion

□ Record completion of the activity in the Exos Work Management System (Mission Control).

Owner: Machine Shop / Exo · Reference documents: Mori Seiki / DMG MORI NLX2500SY/700 Operation & Maintenance Manual (sub-spindle tool-eye sensor-position parameters 2097X / 2098X / 2097Z / 2098Z). Machine record: data/cnc-machines.json (NLX2500, SN NL254140428).